

CASE STUDY 6

AWS Cloud Automation & Serverless Data Processing

Item	Description
Project Type	Cloud Automation / Serverless Data Processing
Role	Software QA Engineer / Python Backend & AWS Cloud Developer
Development Type	Serverless Cloud Application
Programming Language	Python
Development Environment	Visual Studio Code / AWS Management Console / Google Colab
Version Control	Git & GitHub
Project Status	Completed

Project Overview

This project demonstrates the implementation of a serverless cloud-based data processing solution using Amazon Web Services (AWS) and Python.

The solution automatically retrieves multiple CSV datasets from Amazon S3, processes and aggregates the data using Pandas, and uploads the generated analytical results back to Amazon S3.

The project integrates several AWS services including AWS Lambda, Amazon S3, IAM, EventBridge and CloudWatch to demonstrate event-driven automation, secure cloud resource access and serverless data processing.

Throughout the project, emphasis was placed on clean Python development, reusable code organization, cloud architecture, technical documentation and automation best practices.

This project was completed as part of my practical AWS Cloud and Python Automation training and demonstrates my ability to develop structured serverless solutions using modern cloud technologies.

Project Objective

The objectives of this project were:

- Design a serverless cloud solution using AWS.
- Store datasets in Amazon S3.
- Process multiple CSV files automatically.

- Develop reusable Python modules.
 - Implement AWS Lambda functions.
 - Automate cloud workflows.
 - Configure IAM permissions.
 - Monitor execution using CloudWatch.
 - Document cloud architecture.
 - Improve practical AWS development skills.
-

Development Environment

Component	Description
Programming Language	Python
IDE	Visual Studio Code / Google Colab
Cloud Platform	Amazon Web Services (AWS)
Cloud Services	Lambda, S3, IAM, EventBridge, CloudWatch
Operating System	Windows 11
Version Control	Git
Libraries	boto3, pandas
Development Approach	Serverless Cloud Automation

Project Scenarios

The project included implementation of the following practical activities:

- Creating Amazon S3 buckets.
 - Uploading CSV datasets.
 - Reading data from Amazon S3.
 - Processing datasets using Pandas.
 - Aggregating statistical information.
 - Uploading processed files.
 - Executing AWS Lambda functions.
 - Monitoring execution logs.
 - Implementing scheduled automation.
 - Organizing reusable cloud project structure.
-

Development Approach

The project follows a modular serverless architecture based on AWS managed services.

CSV datasets are stored in Amazon S3 and processed using AWS Lambda. The Python application retrieves all available datasets, performs validation and aggregation using Pandas, and uploads the generated analytical report back to Amazon S3.

IAM roles provide secure access between AWS services, while CloudWatch enables execution monitoring and troubleshooting. EventBridge can be used to automate scheduled executions.

The solution emphasizes maintainability, scalability and reusable Python components.

Challenges & Solutions

Several practical challenges were encountered during the project.

- Configuring secure IAM permissions.
 - Managing communication between AWS services.
 - Reading multiple CSV datasets from Amazon S3.
 - Organizing reusable Python modules.
 - Improving project documentation.
 - Designing maintainable serverless architecture.
-

Results

The completed project successfully demonstrated:

- Successful implementation of serverless architecture.
 - Automated CSV processing.
 - Successful integration of Amazon S3 and AWS Lambda.
 - Reliable cloud-based data processing.
 - Practical implementation of boto3 library.
 - Well-structured project documentation.
 - Improved understanding of AWS cloud services.
-

Lessons Learned

The most valuable lessons learned include:

- Serverless architecture simplifies application deployment.
- Amazon S3 provides scalable object storage.
- AWS Lambda enables event-driven automation.
- IAM is essential for secure cloud communication.

- CloudWatch simplifies monitoring and debugging.
- Modular Python code improves maintainability.
- Documentation significantly improves project quality.

Technologies Used

Python • AWS Lambda • Amazon S3 • IAM • EventBridge • CloudWatch • boto3 • Pandas • CSV Processing
• Serverless Computing • Cloud Automation • Git • GitHub

Key Skills Demonstrated

Competency	Demonstrated
Python Programming	✓
AWS Lambda	✓
Amazon S3	✓
boto3	✓
Pandas	✓
Serverless Architecture	✓
Cloud Automation	✓
CSV Processing	✓
IAM	✓
EventBridge	✓
CloudWatch	✓
Git Version Control	✓
Technical Documentation	✓

Project Screenshot

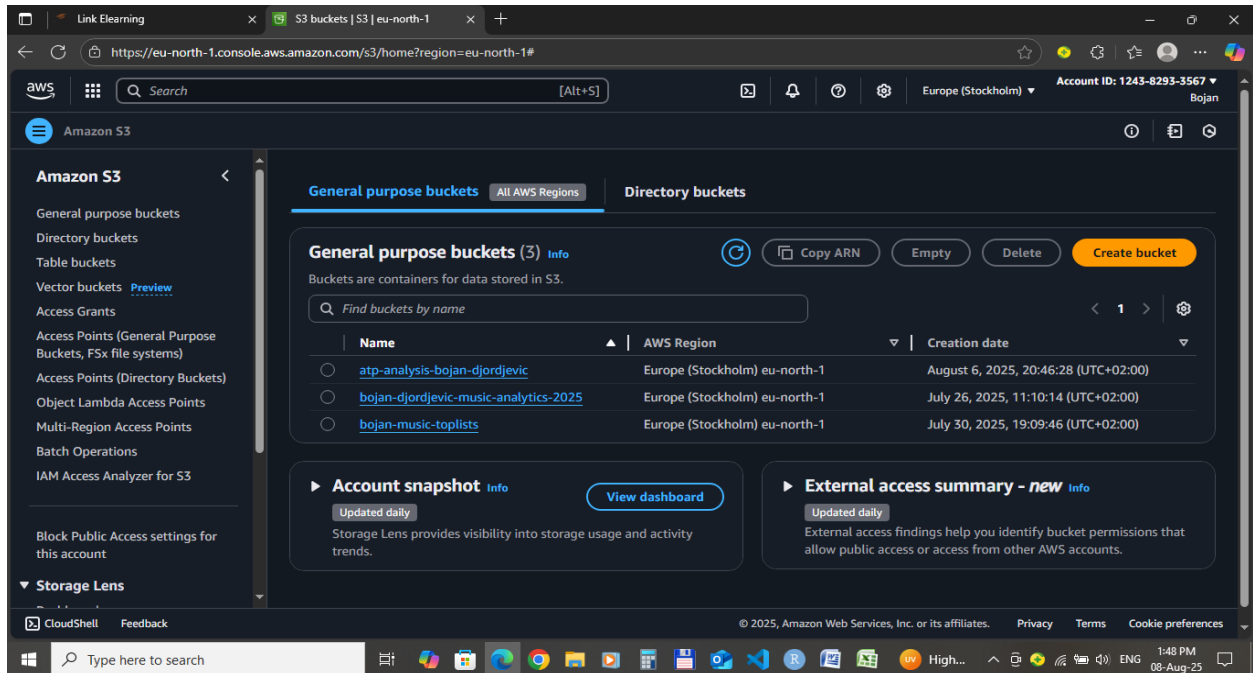


Figure 1 - S3_Bucket_Overview.png

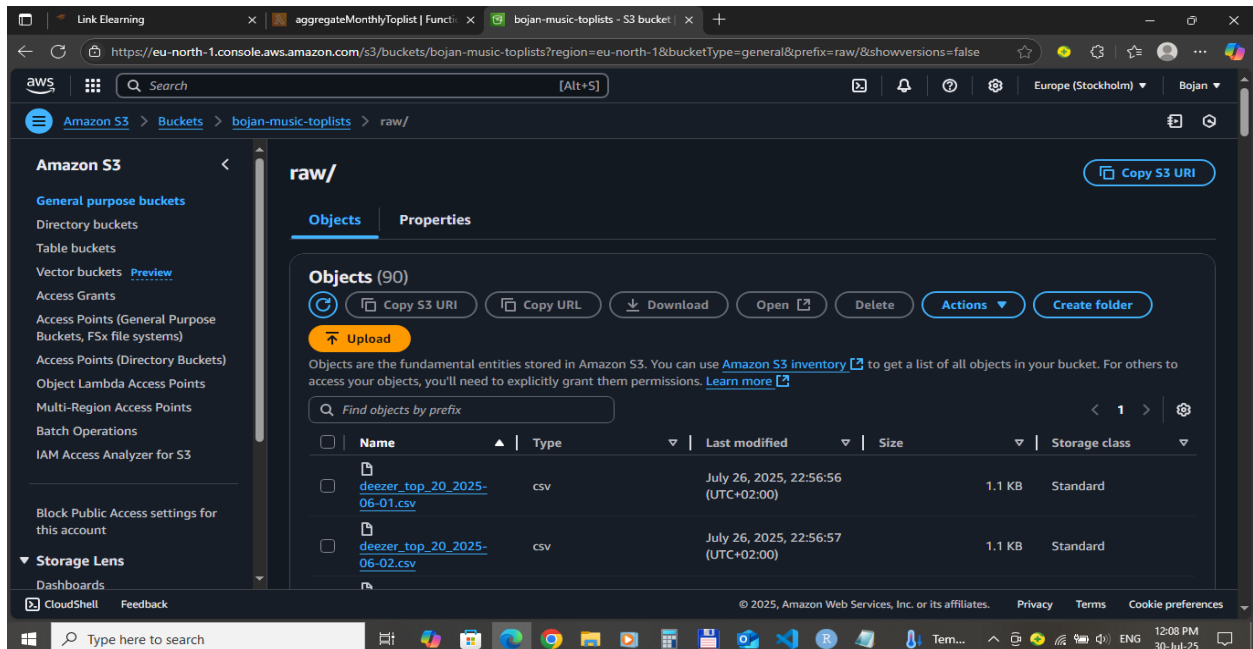


Figure 2 - Raw_CSV_Files.png

A1	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
	Tourname	Date	Series	Court	Surface	Round	Best of	Player_1	Player_2	Winner	Rank_1	Rank_2	Pts_1	Pts_2	Odd_1	Odd_2	Score					
2	Australian	1/3/2000	Internation Outdoor	Hard	1st Round	3	Dosedel S.	Ljubicic I.	Dosedel S.	63	77	-1	-1	-1	-1	-1	6-4 6-2					
3	Australian	1/3/2000	Internation Outdoor	Hard	1st Round	3	Clement A	Enqvist T.	Enqvist T.	56	5	-1	-1	-1	-1	-1	3-6 3-6					
4	Australian	1/3/2000	Internation Outdoor	Hard	1st Round	3	Escude N.	Baccanellc	Escude N.	40	655	-1	-1	-1	-1	-1	6-7 7-5 6-3					
5	Australian	1/3/2000	Internation Outdoor	Hard	1st Round	3	Knippschikl	Federer R.	Federer R.	87	65	-1	-1	-1	-1	-1	1-6 4-6					
6	Australian	1/3/2000	Internation Outdoor	Hard	1st Round	3	Fromberg	Woodbridg	Fromberg	81	198	-1	-1	-1	-1	-1	7-6 5-7 6-4					
7	Australian	1/3/2000	Internation Outdoor	Hard	1st Round	3	Arthurs W.	Gambill J.	Gambill J.	105	58	-1	-1	-1	-1	-1	6-3 6-7 4-6					
8	Australian	1/3/2000	Internation Outdoor	Hard	1st Round	3	Grosjean S	Ille A.	Grosjean S	26	51	-1	-1	-1	-1	-1	6-2 6-1					
9	Australian	1/3/2000	Internation Outdoor	Hard	1st Round	3	Balcells J.	Henman T.	Henman T.	218	11	-1	-1	-1	-1	-1	4-6 6-7					
10	Australian	1/3/2000	Internation Outdoor	Hard	1st Round	3	Hewitt L.	Woodford	Hewitt L.	24	129	-1	-1	-1	-1	-1	6-3 2-6 6-1					
11	Australian	1/3/2000	Internation Outdoor	Hard	1st Round	3	Tebbutt M	Lisnard J.	Lisnard J.	351	134	-1	-1	-1	-1	-1	6-7 7-6 4-6					
12	Australian	1/3/2000	Internation Outdoor	Hard	1st Round	3	Martin A.	Spadea V.	Martin A.	55	21	-1	-1	-1	-1	-1	6-2 6-2					
13	Australian	1/3/2000	Internation Outdoor	Hard	1st Round	3	Drapear S.	Norman M	Norman M	155	15	-1	-1	-1	-1	-1	4-6 4-6					
14	Australian	1/3/2000	Internation Outdoor	Hard	1st Round	3	Petrovic D	Huet S.	Petrovic D	208	122	-1	-1	-1	-1	-1	3-6 6-3 6-3					
15	Australian	1/3/2000	Internation Outdoor	Hard	1st Round	3	Pozzi G.	Sekulov J.	Sekulov J.	79	142	-1	-1	-1	-1	-1	4-6 6-2 2-6					
16	Australian	1/3/2000	Internation Outdoor	Hard	1st Round	3	Stoltenber	Hrbaty D.	Stoltenber	84	20	-1	-1	-1	-1	-1	6-3 6-1					
17	Australian	1/3/2000	Internation Outdoor	Hard	1st Round	3	Goldstein I	Vinck C.	Vinck C.	88	227	-1	-1	-1	-1	-1	6-3 3-6 4-6					
18	Australian	1/3/2000	Internation Outdoor	Hard	2nd Round	3	Enqvist T.	Federer R.	Enqvist T.	5	65	-1	-1	-1	-1	-1	7-6 6-4					
19	Australian	1/3/2000	Internation Outdoor	Hard	2nd Round	3	Dosedel S.	Escude N.	Escude N.	63	40	-1	-1	-1	-1	-1	6-7 4-6					
20	Australian	1/3/2000	Internation Outdoor	Hard	2nd Round	3	Grosjean S	Gambill J.	Grosjean S	26	58	-1	-1	-1	-1	-1	3-6 7-6 6-2					
21	Australian	1/3/2000	Internation Outdoor	Hard	2nd Round	3	Lisnard J.	Henman T.	Henman T.	134	11	-1	-1	-1	-1	-1	6-3 4-6 6-2					
22	Australian	1/3/2000	Internation Outdoor	Hard	2nd Round	3	Hewitt L.	Petrovic D	Hewitt L.	24	208	-1	-1	-1	-1	-1	6-0 6-2					
23	Australian	1/3/2000	Internation Outdoor	Hard	2nd Round	3	Sekulov J.	Martin A.	Martin A.	142	55	-1	-1	-1	-1	-1	6-7 6-7					
24	Australian	1/3/2000	Internation Outdoor	Hard	2nd Round	3	Norman M	Fromberg	Norman M	15	81	-1	-1	-1	-1	-1	7-5 6-4					
25	Australian	1/3/2000	Internation Outdoor	Hard	2nd Round	3	Vinck C.	Stoltenber	Stoltenber	227	84	-1	-1	-1	-1	-1	6-3 4-6 4-6					
26	Australian	1/3/2000	Internation Outdoor	Hard	Quarterfin	3	Enqvist T.	Grosjean S	Enqvist T.	5	26	-1	-1	-1	-1	-1	7-6 7-5					
27	Australian	1/3/2000	Internation Outdoor	Hard	Quarterfin	3	Henman T	Escude N.	Escude N.	11	40	-1	-1	-1	-1	-1	3-6 1-6					
28	Australian	1/3/2000	Internation Outdoor	Hard	Quarterfin	3	Hewitt L.	Stoltenber	Hewitt L.	24	84	-1	-1	-1	-1	-1	6-2 6-2					
29	Australian	1/3/2000	Internation Outdoor	Hard	Quarterfin	3	Martin A.	Norman M	Norman M	55	15	-1	-1	-1	-1	-1	4-6 3-6					
30	Australian	1/3/2000	Internation Outdoor	Hard	Semifinals	3	Enqvist T.	Norman M	Enqvist T.	5	15	-1	-1	-1	-1	-1	3-6 6-2 6-2					
31	Australian	1/3/2000	Internation Outdoor	Hard	Semifinals	3	Escude N.	Hewitt L.	Hewitt L.	40	24	-1	-1	-1	-1	-1	6-3 1-6 4-6					
32	Australian	1/3/2000	Internation Outdoor	Hard	The Final	3	Hewitt L.	Enqvist T.	Hewitt L.	24	5	-1	-1	-1	-1	-1	3-6 6-3 6-2					
33	Gold Flake	1/3/2000	Internation Outdoor	Hard	1st Round	3	Kumar S.	Agenor R.	Agenor R.	1198	94	-1	-1	-1	-1	-1	4-6 4-6					
34	Gold Flake	1/3/2000	Internation Outdoor	Hard	1st Round	3	Behrend T	Popp A.	Behrend T.	116	124	-1	-1	-1	-1	-1	6-4 4-6 6-1					
35	Gold Flake	1/3/2000	Internation Outdoor	Hard	1st Round	3	Rochus C.	Damm M.	Damm M.	120	104	-1	-1	-1	-1	-1	3-6 0-6					

Figure 3 - CSV_Dataset_Preview.png

Link Earning aggregateMonthlyToplist | Function bojan-music-toplists - S3 bucket Top_Artists_June_2025 | Function

https://eu-north-1.console.aws.amazon.com/lambda/home?region=eu-north-1#/functions/Top_Artists_June_2025?tab=configuration

Search [Alt+S]

Lambda > Functions > Top_Artists_June_2025

Code Test Monitor Configuration Aliases Versions

General configuration Info Edit

Description -

Memory 128 MB

Ephemeral storage 512 MB

Timeout 0 min 35 sec

SnapStart None Info

Triggers

Permissions

Destinations

Function URL

Environment variables

Tags

VPC

RDS databases

Monitoring and operations tools

CloudShell Feedback

© 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Type here to search

22°C 1:15 PM 30-Jul-25

Figure 4 - S3_Object_Details.png

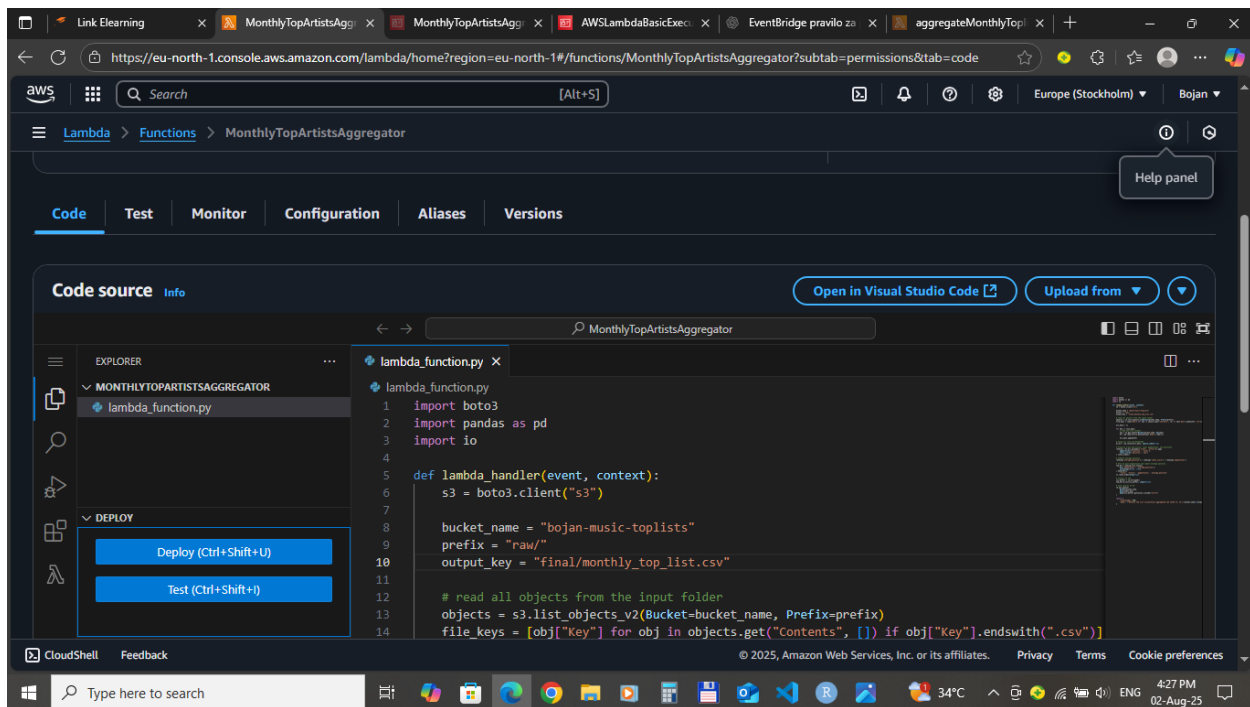


Figure 5 - Lambda_Function.png

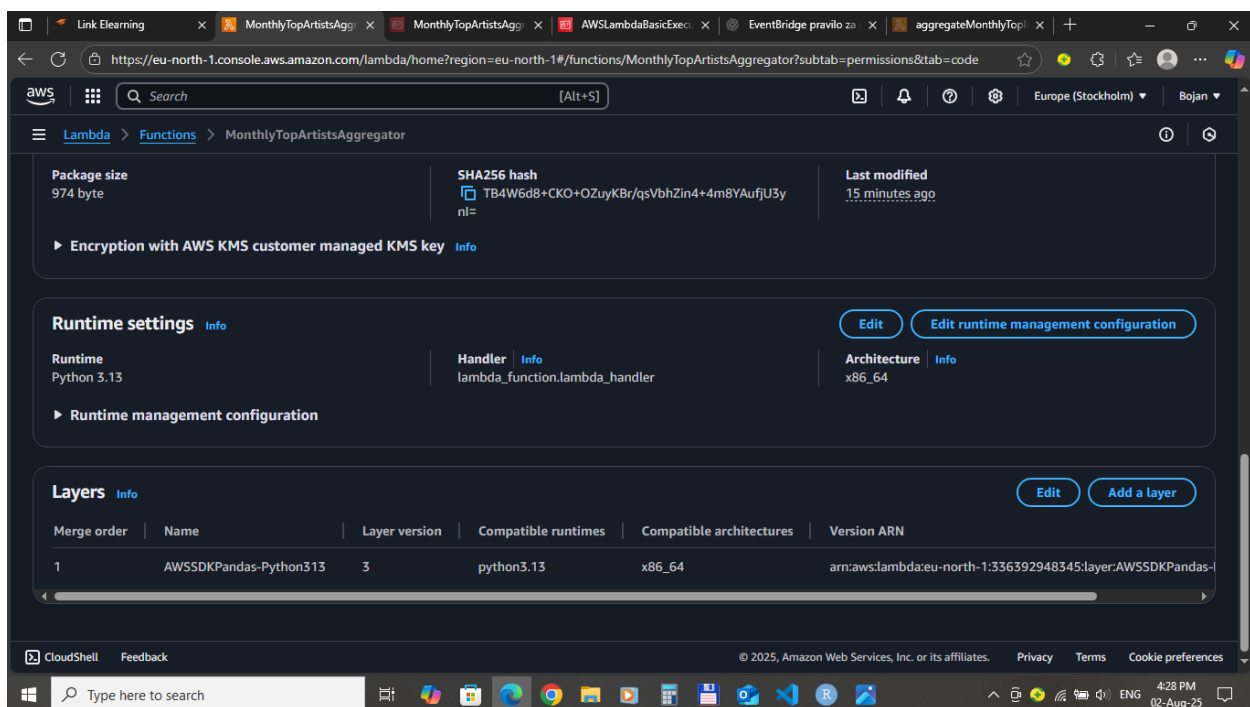


Figure 6 - Lambda_Configuration.png

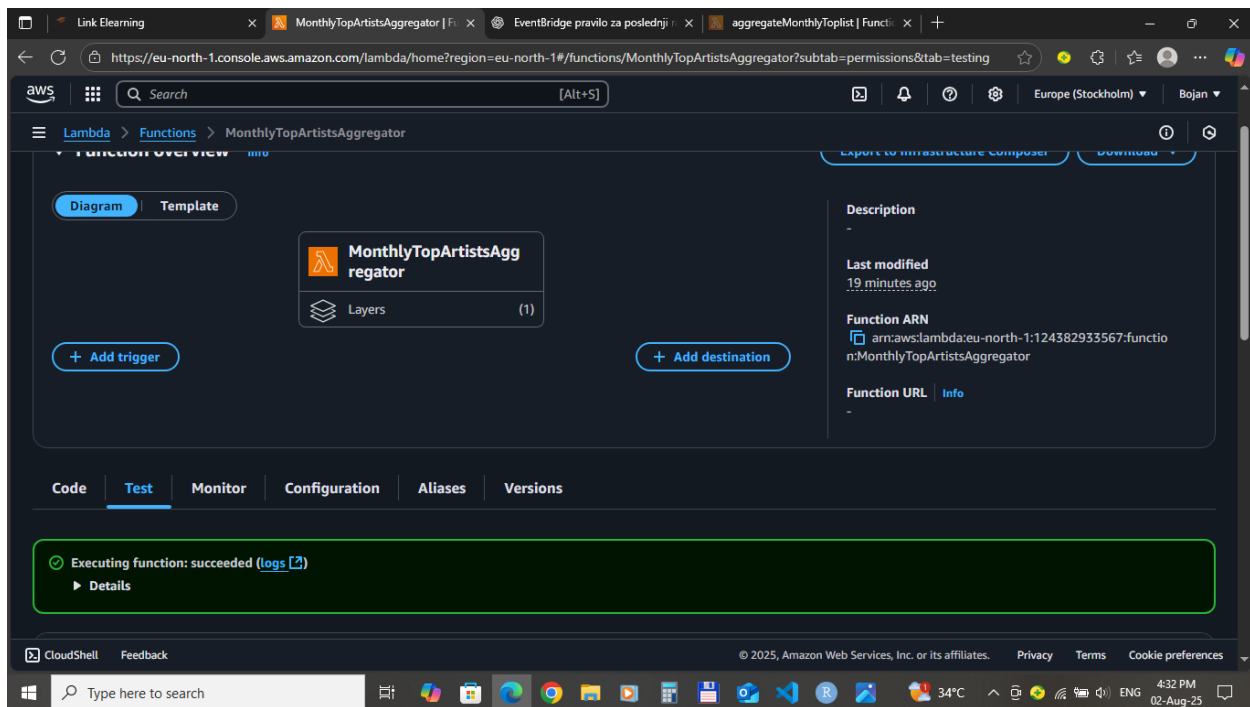


Figure 7 - Lambda_Test_Even.png

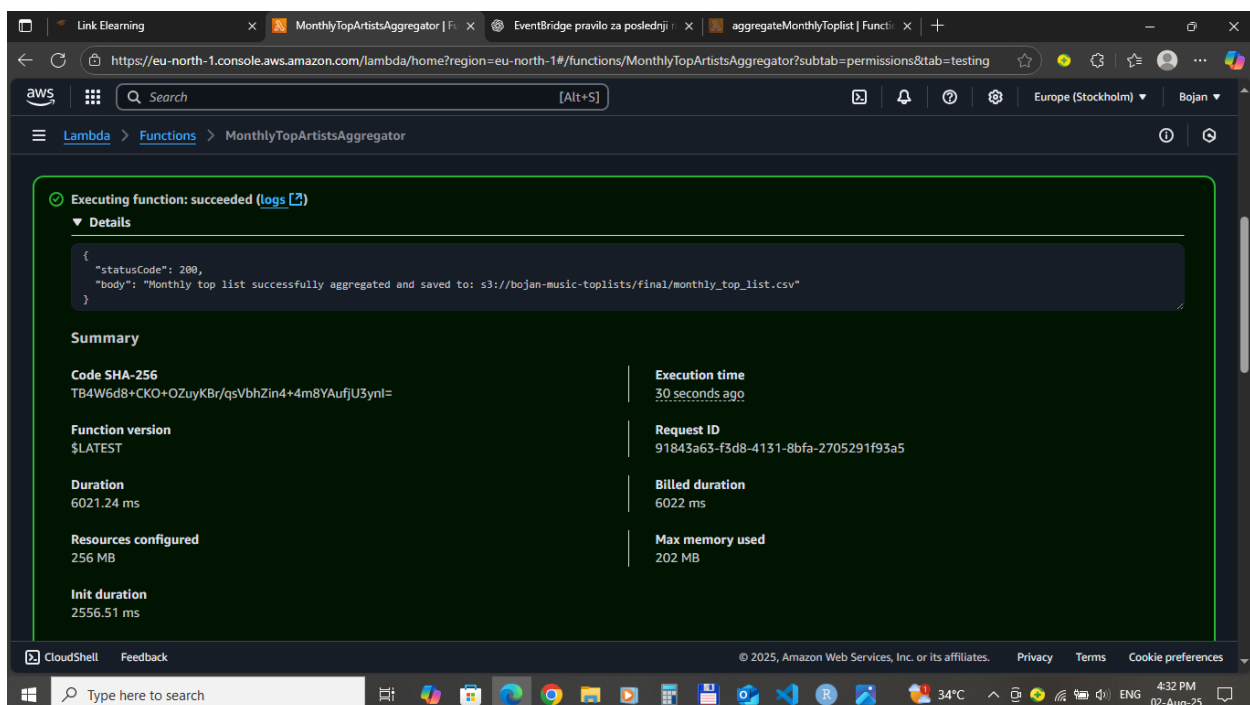


Figure 8 - Lambda_Execution_Result.png

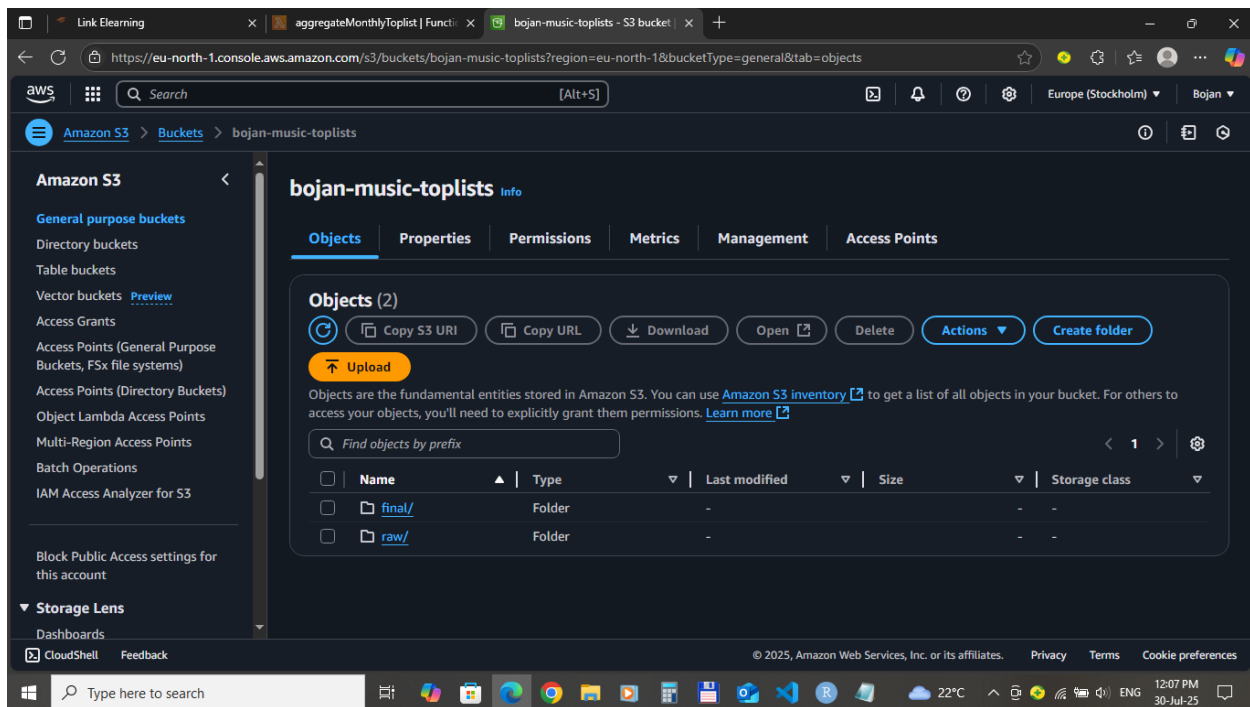


Figure 9 - Final_Output_File.png

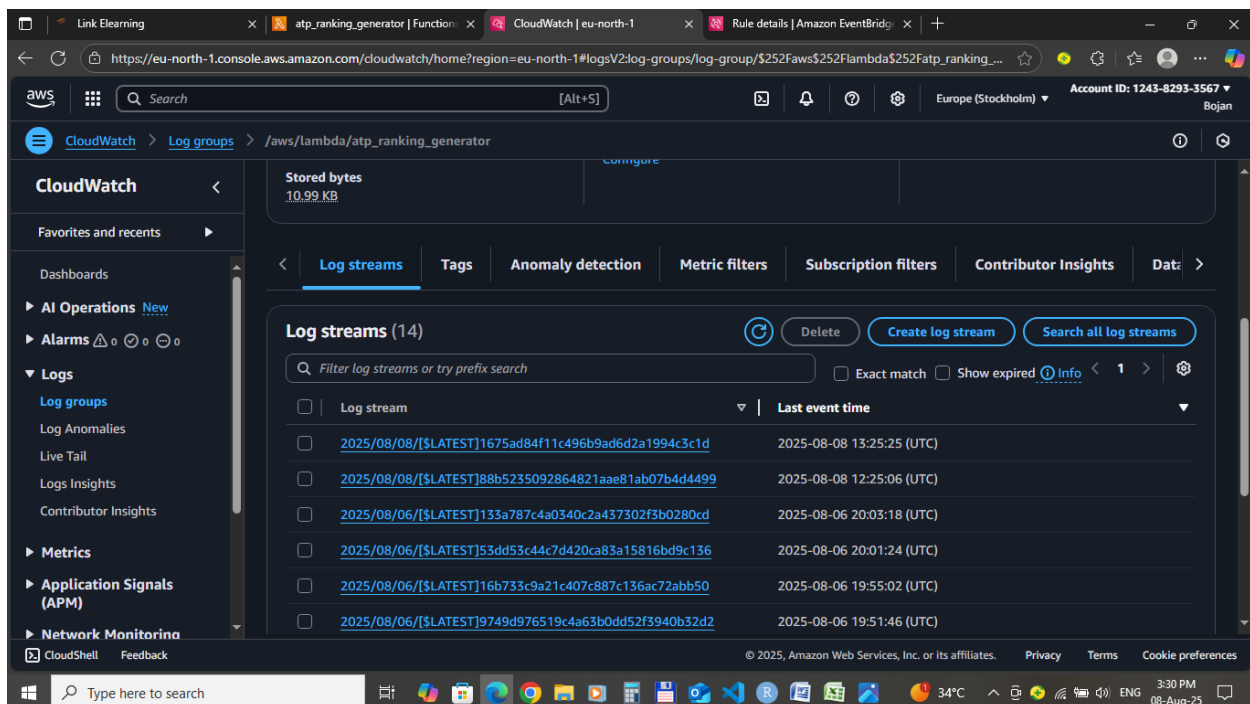


Figure 10 - CloudWatch_Logs.png

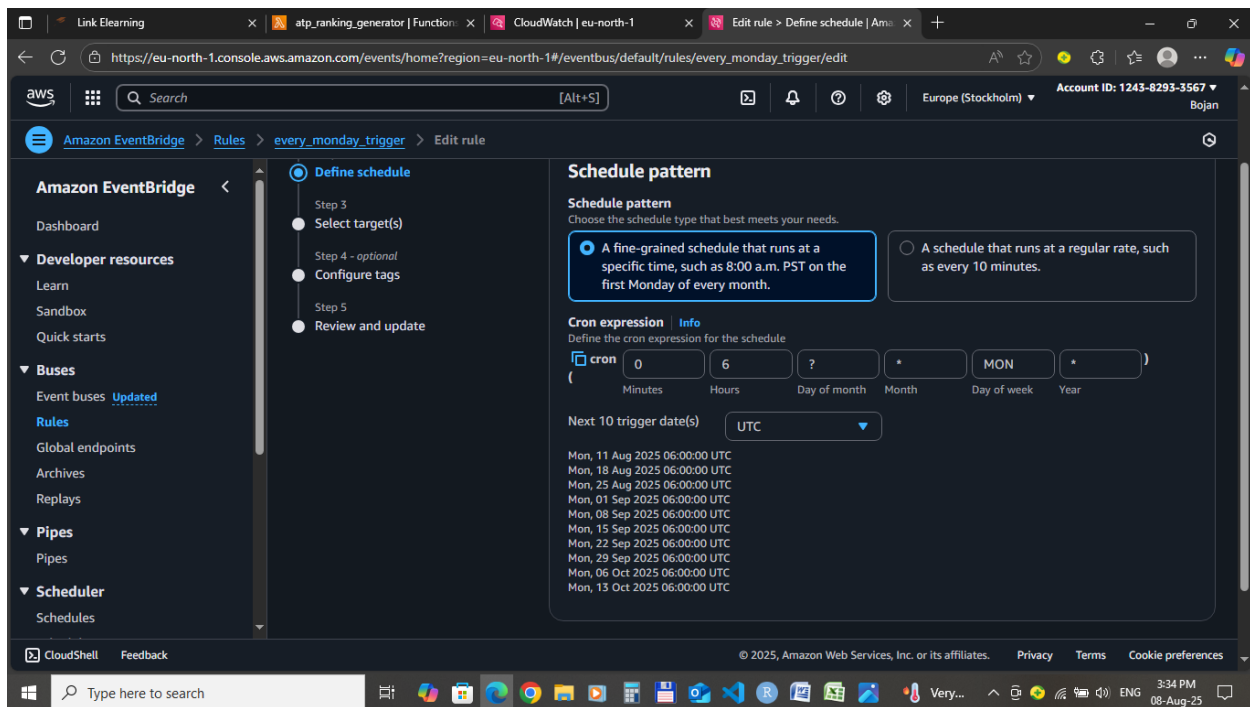


Figure 11 - EventBridge_Rule.png

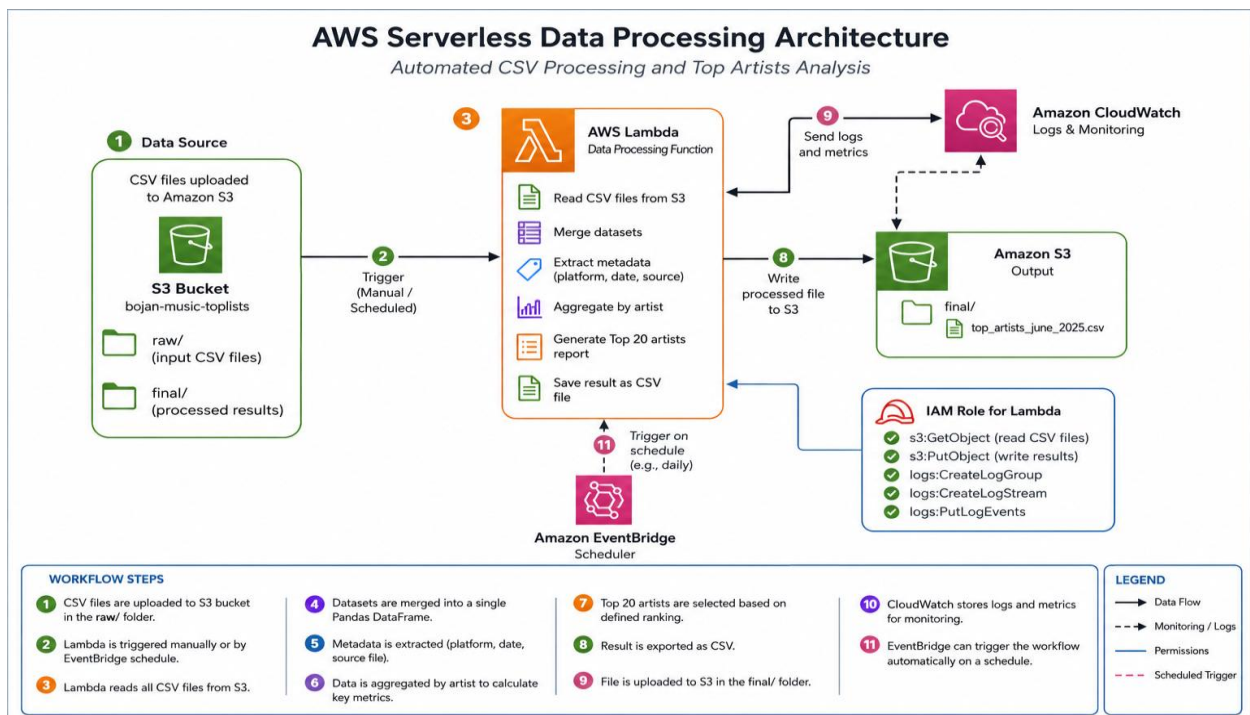


Figure 12 - Project_Architecture.png

GitHub Repository

[View **AWS Cloud Automation & Serverless Data Processing** Project on GitHub](#)

Recruiter Summary

This project demonstrates practical experience in designing and implementing serverless cloud solutions using Amazon Web Services and Python. It highlights my ability to integrate multiple AWS services, automate cloud workflows, process structured datasets and develop maintainable Python applications. The project reflects a structured engineering approach, combining cloud infrastructure, backend development and technical documentation into a complete real-world solution.
